

Response to Reviewers

1. General Response

We thank the Associate Editor and all reviewers for the careful second-round assessment. Three concerns recurred across the reports. These were the theoretical basis of the functional decoupling, the absence of statistical validation, and evaluation on a single benchmark. Each is answered with new analysis or new experiments in the manuscript. The table below maps every concern to the change and its location.

Concern	How we address it	Where
Theory of the decoupling (R2, R4)	Exact entropy identity replacing the previous approximate risk decomposition, one derivation per module, and a 2×2 ambiguity-stratified analysis testing their prediction	Sec. 3.5, and Sec. 4.3.2 with Table 7
Statistical validation (R2, R6)	Five seeds, 95% t -intervals, paired t -tests against every matched-protocol comparator with Holm correction	Sec. 4.2.1 and Table 2, with full statistics in Table R1 of this letter
Single benchmark, dataset scale (R2, R4)	Cross-dataset study: a larger EMOTIC benchmark, plus the Flickr-LDL and Emotion6 label-distribution benchmarks	Sec. 4.5, Tables 15 and 16
Recent baselines (R4, EiC)	Two 2026 intent methods added and compared, one under a matched backbone, and four recent MLLMs evaluated zero-shot	Sec. 2, Tables 2, 3, 14
Abstract comparability (AE)	More intuitive presentation of advantages	Abstract
Proofreading, bibliography (R4, EiC)	Venue fields corrected, new works discussed individually, 52 items	Bibliography, Sec. 2

Two points of context apply throughout, so we state them once here rather than repeating them under each comment.

The cross-dataset study. It now spans two axes, and both are in the manuscript. EMOTIC is the larger benchmark (23,571 images, 34,320 annotated instances) and keeps our multi-label output geometry, but releases multi-annotator labels only for validation and test, so we report it as a controlled cross-validation on that multi-annotated pool rather than as a leaderboard comparison. Flickr-LDL and Emotion6 are smaller yet carry native subjective label distributions under standard published protocols. Together they answer both the scale and the breadth questions.

What this letter adds. The manuscript is at the 35-page limit, so three items live here instead. These are the complete per-method significance statistics (Table R1), the seed dispersion behind the cross-dataset table (Table R2), and the per-class F1 attribution behind the manuscript’s one-sentence interpretability summary (Table R3). Everything else we describe below is in the revised manuscript, marked in red.

2. Response to Reviewer 2

We thank the reviewer for acknowledging the improvements in methodological detail, evaluation, and presentation, and for stating the remaining concerns so precisely. All four are addressed with changes in the manuscript, summarized below.

Comment 1

There is still a lack of theoretical justification of functionally separate two types of ambiguity.

Response: *Sec. 3.5 is rebuilt around an exact identity plus one derivation per module, and Sec. 4.3.2 adds an experiment that tests what they predict.*

We rebuild the analysis on an exact footing. Sec. 3.5 now derives the three-term identity $H(Y_c | B) = H(Y_c | X) + I(Y_c; X | Z) + I(Y_c; Z | B)$ (Eq. (15)), which separates irreducible annotator disagreement, label-relevant information the frozen representation discards, and information the baseline readout leaves unused. The middle term is unrecoverable downstream, and since our text prior is itself computed from the same frozen feature it cannot address it; what SLR-C recovers is bounded by the third term. Stating the decomposition this way removes an inconsistency in the previous two-term version, which could be read as claiming that a module reading the same representation recovers what that representation had already discarded. Each module is then characterized on its own terms. SLR-C is stated through the conditional mutual information it supplies (Eq. (16)) and the margin it shifts inside the screened candidate set (Eq. (17)). UTD is stated as learning using privileged information, whose mean-squared-error-optimal weight decreases in annotator agreement and so motivates the gate $1 - \omega$ (Eq. (18)).

It establishes that the three terms have distinct origins, namely the subjectivity of the annotation, the frozen representation map, and the readout, which motivates treating them with distinct mechanisms; whether that separation helps in practice is an empirical question, and under the unified controls we test (Table 6) the decoupled design is better. We are careful not to claim that both ambiguities can be reduced: the irreducible term admits no reduction by any learner, and UTD’s role is to limit target-estimation error and avoid overfitting unreliable hard targets, not to lower it. Whether that pays off is empirical, so we add an experiment to test it. The new 2×2 ambiguity-stratified analysis below confirms the prediction on the semantic axis, where SLR-C gains only in the high-semantic-ambiguity cells, and shows UTD inert wherever annotator agreement is high, with its only significant gain where both ambiguities coincide. The full model does not dominate the doubly-ambiguous cell, so we claim the predicted localization of the two effects rather than a gain from composing them.

Where and what we changed. the derivations (15)–(18) and the module

mapping in Sec. 3.5, the 2×2 ambiguity-stratified analysis in Sec. 4.3.2 and Table 7, and the unified-versus-decoupled control in Table 6.

Comment 2

In spite of the inclusion of new experiments, statistical validation is still missing. Confidence intervals, hypothesis testing, and robustness to multiple training runs are necessary to prove that reported improvements are statistically significant.

Response: *All three elements are now in place, with the complete per-method statistics in Table R1 of this letter.*

We extend the main experiments from three to **five** independent seeds, re-training FDIL and every matched-protocol method on each seed. Table 2 now reports 95% t -intervals instead of bare standard deviations, and Sec. 4.2.1 reports paired two-sided t -tests against each matched-protocol comparator, paired on the training run and Holm-corrected within the family of methods compared on that metric. The published-only rows (CoT4Intent, HVU-CLIP, IntentMLM) are quoted from their papers and are therefore not part of the paired tests, which we state explicitly. This is complemented by the sample-level paired bootstrap behind Table 11, so run-to-run and sample-to-sample variance are both quantified.

Avg. F1, mAP, and hard-subset F1 are significant against every matched-protocol comparator after correction, while only Samples F1 is not against our own UTD-only ablation ($p_H = 0.0755$). The manuscript summarizes the reliable gains, and the full statistics are provided in Table R1.

Where and what we changed. Sec. 4.2.1, the confidence intervals in Table 2, the paired-bootstrap discussion accompanying Table 11, and Table R1 of this letter.

[Manuscript Table 7] 2×2 ambiguity-stratified analysis on cross-fitted Intentionomy train predictions. Cells give Δ per-sample F1 against the shared baseline; * marks Holm-corrected $p < 0.05$.

		Low supervisory ($\omega \geq 2/3$)		High supervisory ($\omega = 1/3$)	
		Low sem.	High sem.	Low sem.	High sem.
	n	1,271	588	5,099	5,781
Base	F1	72.23	41.56	40.54	22.90
	+SLR-C	-0.43	+4.08*	+0.49	+1.17*
Δ F1	+UTD	-0.05	-0.07	+0.03	+0.84*
	Full FDIL	-0.71	+2.18	+0.15	+1.14*

Table R1: Paired t -test of FDIL against each matched-protocol method over five seeds, for all six reported metrics. Δ is the mean per-seed difference in points; CI_{95} is its 95% t -interval; p_H is Holm-corrected across the family of compared methods within each metric. The manuscript summarizes this table in Sec. 4.2.1 and carries the corresponding intervals in Table 2.

Metric	Method	Δ	CI_{95}	t	p	p_H
Macro F1	CLIP ViT-L/14	+4.57	[3.37, 5.76]	10.62	<0.001	0.0014
	UTD only (ours)	+2.00	[0.10, 3.91]	2.92	0.0434	0.0434
	HLEG	+4.52	[3.40, 5.63]	11.24	<0.001	0.0014
	LabCR	+4.03	[2.86, 5.19]	9.60	<0.001	0.0014
	IntCLIP	+6.92	[5.84, 7.99]	17.81	<0.001	<0.001
Micro F1	CLIP ViT-L/14	+4.92	[2.20, 7.64]	5.02	0.0074	0.0148
	UTD only (ours)	+1.88	[0.35, 3.41]	3.42	0.0268	0.0268
	HLEG	+5.05	[3.27, 6.83]	7.89	0.0014	0.0042
	LabCR	+3.12	[2.40, 3.83]	12.13	<0.001	0.0011
	IntCLIP	+5.16	[4.29, 6.03]	16.48	<0.001	<0.001
Samples F1	CLIP ViT-L/14	+4.29	[1.95, 6.64]	5.09	0.0070	0.0141
	UTD only (ours)	+1.60	[-0.26, 3.47]	2.39	0.0755	0.0755
	HLEG	+4.75	[3.18, 6.31]	8.42	0.0011	0.0044
	LabCR	+2.80	[1.54, 4.06]	6.17	0.0035	0.0105
	IntCLIP	+5.30	[4.07, 6.54]	11.91	<0.001	0.0014
Avg. F1	CLIP ViT-L/14	+4.59	[2.88, 6.31]	7.43	0.0018	0.0035
	UTD only (ours)	+1.83	[0.11, 3.55]	2.96	0.0416	0.0416
	HLEG	+4.77	[3.39, 6.16]	9.55	<0.001	0.0020
	LabCR	+3.32	[2.46, 4.17]	10.79	<0.001	0.0017
	IntCLIP	+5.79	[4.86, 6.73]	17.26	<0.001	<0.001
mAP	CLIP ViT-L/14	+3.76	[2.93, 4.59]	12.58	<0.001	<0.001
	UTD only (ours)	+1.64	[0.98, 2.30]	6.94	0.0023	0.0023
	HLEG	+4.11	[3.55, 4.68]	20.22	<0.001	<0.001
	LabCR	+3.41	[2.48, 4.35]	10.16	<0.001	0.0011
	IntCLIP	+7.12	[6.80, 7.44]	61.60	<0.001	<0.001
Hard F1	CLIP ViT-L/14	+7.14	[4.73, 9.56]	8.21	0.0012	0.0036
	UTD only (ours)	+4.15	[0.93, 7.37]	3.58	0.0232	0.0232
	HLEG	+8.47	[6.18, 10.75]	10.28	<0.001	0.0020
	LabCR	+7.67	[4.85, 10.48]	7.56	0.0016	0.0036
	IntCLIP	+10.51	[8.29, 12.73]	13.13	<0.001	0.0010

Comment 3

Moreover, the evaluation is limited to the single benchmark dataset that makes it impossible to assess generalization ability of the approach.

Response: We agree, and the manuscript now carries a cross-dataset study on the larger EMOTIC benchmark and on two label-distribution benchmarks.

Sec. 4.5 adds two kinds of transfer. Sec. 4.5.1 evaluates the larger EMOTIC benchmark under a controlled 5×5 cross-validation on its multi-annotated pool, with five seed-level paired observations, where every aggregate metric improves significantly over the matched baseline. Sec. 4.5.2 then evaluates Flickr-LDL (11,150 images) and Emotion6 (1,980 images). In our implementation, each uses its untouched official test split over five seeds with paired t -tests. These vary a different axis than a further multi-label dataset would. Label distribution learning replaces independent per-class decisions with a point on the probability simplex, so the threshold search disappears and only the two functional roles remain.

Results. On Flickr-LDL, FDIL significantly improves rank correlation and the divisiveness error but significantly degrades Clark distance. On Emotion6

the only reliable change is Clark distance, the very metric it loses on Flickr-LDL. The margins are small and Emotion6’s 594-image test split has little power at this effect size, so we do not present either dataset as a new state of the art for label distribution learning. The manuscript table also reports four published label-distribution objectives, δ -LDL [24], LDL-LRR [22], LDL-DPA [23], and LDL-DVS [25], run at their default weights on identical features, splits, seeds, and head. All four stay within 0.001 of the matched baseline on every metric except μ , which shifts by at most 0.09 (Table 16), which also limits what those comparisons can establish.

What we do take from the study. The division of labour replicates on both datasets. SLR-C primarily affects the pointwise-fit metrics (KLD, cosine, μ), while gated UTD moves rank correlation and leaves KLD no better. That separation holds where the two modules cannot share a fitted threshold, which is the most plausible way it could otherwise have been an artefact. Table R2 below gives the seed dispersion and the per-metric p -values that the manuscript table omits for space.

Where and what we changed. the new Sec. 4.5 with Tables 15 and 16, the dataset descriptions and label-distribution metrics in Sec. 4, and the added transfer limitations in Sec. 5.

Table R2: Full label-distribution comparison behind manuscript Table 16: untouched official test splits, mean $\pm$ standard deviation over five seeds. δ -LDL [24], LDL-LRR [22], LDL-DPA [23], and LDL-DVS [25] are published label-distribution objectives run at their default weights on identical features, splits, seeds, and prediction head. Bold marks the best entry per metric within each block. The last row of each block gives Δ and the paired t -test p for FDIL against the matched baseline over the five paired runs, with $p < 0.05$ in italics.

Method	Cheby. $\downarrow$	Clark $\downarrow$	KLD $\downarrow$	Cos. $\uparrow$	Spear. $\uparrow$	$\mu(\%) \uparrow$	DVSE $\downarrow$
<i>Flickr-LDL</i> (2,229 test images)							
Baseline	0.2039 $\pm$ 0.0012	2.3033 $\pm$ 0.0044	0.3433 $\pm$ 0.0016	0.8966 $\pm$ 0.0007	0.7109 $\pm$ 0.0026	71.40 $\pm$ 0.14	0.0772 $\pm$ 0.0020
δ -LDL	0.2039 $\pm$ 0.0014	2.3032 $\pm$ 0.0046	0.3434 $\pm$ 0.0018	0.8966 $\pm$ 0.0008	0.7109 $\pm$ 0.0026	71.39 $\pm$ 0.16	0.0772 $\pm$ 0.0020
LDL-LRR	0.2039 $\pm$ 0.0012	2.3033 $\pm$ 0.0044	0.3433 $\pm$ 0.0016	0.8966 $\pm$ 0.0007	0.7109 $\pm$ 0.0026	71.40 $\pm$ 0.14	0.0772 $\pm$ 0.0020
LDL-DPA	0.2039 $\pm$ 0.0012	2.3033 $\pm$ 0.0044	0.3433 $\pm$ 0.0016	0.8966 $\pm$ 0.0007	0.7109 $\pm$ 0.0026	71.40 $\pm$ 0.14	0.0772 $\pm$ 0.0020
LDL-DVS	0.2039 $\pm$ 0.0012	2.3034 $\pm$ 0.0044	0.3433 $\pm$ 0.0016	0.8966 $\pm$ 0.0007	0.7108 $\pm$ 0.0026	71.40 $\pm$ 0.14	0.0772 $\pm$ 0.0020
+ SLR-C	0.2031 $\pm$ 0.0006	2.3031 $\pm$ 0.0024	0.3406 $\pm$ 0.0010	0.8973 $\pm$ 0.0007	0.7129 $\pm$ 0.0010	71.58 $\pm$ 0.09	0.0764 $\pm$ 0.0014
+ UTD	0.2026 $\pm$ 0.0011	2.3123 $\pm$ 0.0025	0.3454 $\pm$ 0.0029	0.8961 $\pm$ 0.0013	0.7137 $\pm$ 0.0012	71.31 $\pm$ 0.18	0.0729 $\pm$ 0.0011
FDIL	0.2027 $\pm$ 0.0007	2.3113 $\pm$ 0.0027	0.3433 $\pm$ 0.0016	0.8967 $\pm$ 0.0006	0.7141 $\pm$ 0.0012	71.45 $\pm$ 0.11	0.0738 $\pm$ 0.0009
Δ , p (FDIL)	-0.0011, .178	+0.0079, .008	+0.0000, .978	+0.0001, .874	+0.0032, .017	+0.05, .542	-0.0034, .009
<i>Emotion6</i> (594 test images)							
Baseline	0.1794 $\pm$ 0.0019	1.5979 $\pm$ 0.0025	0.2532 $\pm$ 0.0026	0.8947 $\pm$ 0.0014	0.7223 $\pm$ 0.0025	64.20 $\pm$ 0.37	0.0784 $\pm$ 0.0009
δ -LDL	0.1798 $\pm$ 0.0020	1.5979 $\pm$ 0.0026	0.2538 $\pm$ 0.0028	0.8943 $\pm$ 0.0015	0.7216 $\pm$ 0.0029	64.11 $\pm$ 0.40	0.0785 $\pm$ 0.0009
LDL-LRR	0.1794 $\pm$ 0.0019	1.5979 $\pm$ 0.0026	0.2532 $\pm$ 0.0026	0.8947 $\pm$ 0.0014	0.7223 $\pm$ 0.0025	64.20 $\pm$ 0.37	0.0784 $\pm$ 0.0009
LDL-DPA	0.1794 $\pm$ 0.0019	1.5979 $\pm$ 0.0025	0.2532 $\pm$ 0.0026	0.8947 $\pm$ 0.0014	0.7223 $\pm$ 0.0025	64.20 $\pm$ 0.37	0.0784 $\pm$ 0.0009
LDL-DVS	0.1794 $\pm$ 0.0019	1.5981 $\pm$ 0.0026	0.2532 $\pm$ 0.0026	0.8946 $\pm$ 0.0014	0.7222 $\pm$ 0.0024	64.20 $\pm$ 0.37	0.0783 $\pm$ 0.0009
+ SLR-C	0.1797 $\pm$ 0.0016	1.5976 $\pm$ 0.0053	0.2515 $\pm$ 0.0019	0.8955 $\pm$ 0.0013	0.7231 $\pm$ 0.0026	64.52 $\pm$ 0.28	0.0781 $\pm$ 0.0006
+ UTD	0.1804 $\pm$ 0.0018	1.5949 $\pm$ 0.0047	0.2537 $\pm$ 0.0033	0.8941 $\pm$ 0.0018	0.7256 $\pm$ 0.0026	64.12 $\pm$ 0.43	0.0791 $\pm$ 0.0014
FDIL	0.1802 $\pm$ 0.0011	1.5910 $\pm$ 0.0023	0.2513 $\pm$ 0.0006	0.8953 $\pm$ 0.0008	0.7219 $\pm$ 0.0026	64.41 $\pm$ 0.12	0.0781 $\pm$ 0.0009
Δ , p (FDIL)	+0.0008, .438	-0.0069, .013	-0.0019, .190	+0.0006, .439	-0.0003, .740	+0.21, .304	-0.0003, .683

Comment 4

The analysis of computational efficiency, visualization of failures, and discussion of deployment are needed to make the contribution stronger.

Response: All three are in the manuscript, and we reproduce the efficiency table and the failure case below.

Efficiency. Table 17 reports inference-active parameters, per-image FLOPs, and batch-1 latency under one shared protocol, at 2.41M parameters and 0.47 ms per image for full FDIL, with UTD-only at exactly baseline cost. **The table now also covers LabCR, the strongest prior method under our matched protocol, whose label-query Transformer costs 153.29M parameters and 1.395 ms per image.**

Failure visualization. Fig. 5, reproduced below, shows both a success and a genuine failure case, and the interpretability analysis reports that only two of the 28 classes regress. The per-class attribution behind that sentence is condensed out of the manuscript for space, so we reproduce it here (Table R3). The larger of the two regressions is *CreativeUnique* (−5.3), a broad category overlapping several neighbouring design- and art-related intents that the residual student occasionally resolves toward a competing neighbour.

Deployment. The efficiency discussion notes that the UTD branch can be deployed alone at zero additional inference cost, since the text teacher is discarded after training, and the Limitations paragraph records the offline text-generation cost any deployment must budget for.

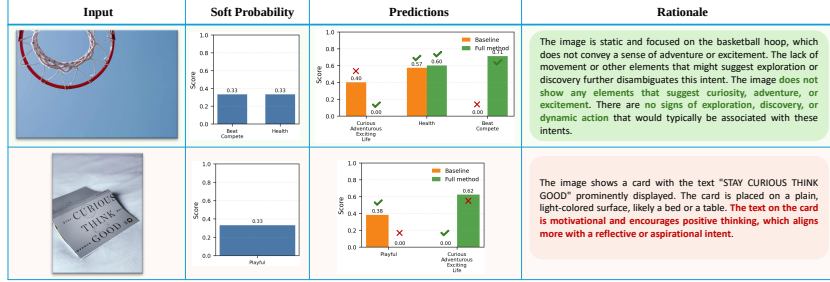
Where and what we changed. Table 17 and its surrounding discussion, the Interpretability Analysis subsection and Fig. 5, and the deployment and preparation-cost notes in Sec. 5.

[Manuscript Table 17] Feature-level inference cost on pre-extracted CLIP ViT-L/14 features: inference-active parameters, per-image FLOPs, and batch-1 latency on an RTX 4090. The shared frozen encoder is excluded.

Method	Params (M)	FLOPs (M)	Latency (ms)
Baseline	1.20	2.40	0.109
IntCLIP [9]	0.02	22.02	0.111
LabCR [8]	153.29	11 526.8	1.395
HLEG [7]	203.70	12 075.9	2.725
FDIL (UTD only)	1.20	2.40	0.109
FDIL	2.41	4.93	0.466

3. Response to Reviewer 3 (Associate Editor)

We thank the Associate Editor for clarifying the intent of the previous comment. The Abstract is now built around comparative statements, as requested.



[Manuscript Fig. 5] Qualitative case study on representative Intentionomy samples. Top: the baseline predicts an incorrect alternative and FDIL rejects that distractor using counterfactual rationale evidence. Bottom: a genuine failure of FDIL, on a label with only 0.33 annotator agreement, where its alternative remains semantically plausible.

Table R3: Per-class F1 attribution of FDIL versus the CLIP baseline (matched class-wise protocol, representative seed). Top: five largest improvements. Bottom: the only two regressions among all 28 classes. The manuscript retains its one-sentence summary in the Interpretability Analysis.

Intent class	Subset	Baseline	FDIL	Δ
WorkILike	Hard	14.6	43.0	+28.5
PassionAboutSomething	Hard	11.3	28.6	+17.3
SocialLifeFriendship	Hard	33.9	51.2	+17.2
ManageableMakePlan	Medium	17.5	32.0	+14.5
BeatCompete	Medium	54.6	68.1	+13.5
FineDesignLearnArt-Arch	Easy	84.0	83.1	-1.0
CreativeUnique	Medium	41.9	36.6	-5.3

Comment 1

Although the manuscript has been revised in response to the previous reviewer comments, the Abstract still does not fully address the main concern. My previous comment was not intended to encourage the authors to simply present more numerical values. Rather, I expected a quantitative comparison with existing methods, such as clearly stating the percentage improvement over the baseline. Presenting objective comparative metrics is essential for highlighting the significance of the proposed method and attracting readers' interest. Therefore, I believe this part still requires further revision.

Response: *The Abstract now states relative gains over the strongest prior result rather than absolute scores.*

We agree that a relative comparison conveys the size of the advance in a way absolute scores do not. The Abstract now states an explicit relative gain over the strongest prior result under our matched protocol, and reads “On Intentionomy, FDIL improves over the strongest prior method by 6.2% relative average F1 and 28.4% relative hard-subset F1.”

Both percentages are relative to LabCR, the strongest prior method under our matched protocol: 57.28 against 53.96 Avg. F1 in Table 2, and 34.66 against 26.99 hard-subset F1 in Table 4, which we extend with the reproduced matched-protocol rows so that both percentages are now checkable in the manuscript itself (+3.32 Avg. F1 and +7.67 hard-subset F1, the latter significant at $p_H = 0.004$). The claim is scoped to the methods we reproduced under the matched protocol rather than to all comparators, since only those rows share our features, split, and thresholding and can therefore support a paired test.

The surrounding text then summarizes the cross-dataset evidence on EMOTIC, Flickr-LDL, and Emotion6 without adding a further list of isolated metric values. This keeps the Abstract focused on objective comparative gains, as requested, and leaves the complete per-method results and five-seed confidence intervals to Table 2.

Where and what we changed. the rewritten quantitative-comparison sentences in the Abstract.

4. Response to Reviewer 4

We thank the reviewer for the continued scrutiny. The dataset-scale and recent-baseline requests are answered with new experiments, the theoretical request with a rebuilt analysis in the information-theoretic and feature-selection terms the reviewer names, and the proofreading item with a bibliography-wide pass.

Comment 1

There must be larger-scale datasets to better evaluate the proposed methods.

Response: *We agree, and the manuscript now evaluates the larger EMOTIC benchmark, together with two label-distribution benchmarks that broaden the evaluation along a second axis. EMOTIC provides a larger transfer benchmark, although the ambiguity-aware evaluation is necessarily restricted to its 10,613-person multi-annotated pool.*

Scale. Sec. 4.5.1 adds EMOTIC [29], with 23,571 images and 34,320 annotated instances, which keeps the multi-label output geometry our method is defined on. Because multi-annotator labels are released only for validation and test, the ambiguity-aware evaluation is confined to that multi-annotated pool of 10,613 persons, on which we run a controlled 5×5 cross-validation. Each seed aggregates its complete five-fold out-of-fold performance over the pool, so a seed is one paired observation and every test uses five paired values. Full FDIL improves the baseline by +0.42 Macro, +0.67 Micro, +0.62 Samples F1 and +0.38 mAP, each significant across seeds at $p \leq 0.005$, with the same division of labour, since gated UTD carries the thresholded F1 metrics and SLR-

C carries mAP. We report this as a controlled transfer analysis on a held-out multi-annotated pool, not as a comparison against published EMOTIC numbers trained on the single-annotator training partition. Only SLR-C’s candidate size and prior weight are retuned for the new label space ($K = 26$, $\alpha = 0.09$). As an additional check reported here, nested cross-validation of this selection gives 42.69 mAP, essentially reproducing the reported SLR-C result.

Breadth. Sec. 4.5.2 then evaluates Flickr-LDL (11,150 images, per-image annotator vote distributions over eight emotions) and Emotion6 (1,980 images, seven categories including neutral). Neither is individually larger than Intentionomy, and we do not present them as answering the scale request. They answer evaluation breadth, because label distribution learning removes the class-wise threshold search entirely, leaving only the two functional roles to carry the transfer. The scale and the breadth axes are therefore covered by different datasets. In our implementation, each uses its untouched official test split, five seeds with paired t -tests, and out-of-fold teacher predictions so no training target derives from an evaluated sample. On Flickr-LDL, FDIL significantly improves rank correlation and the divisiveness error but significantly degrades Clark distance, and on Emotion6 the only reliable change is Clark distance, the metric it loses on Flickr-LDL. We do not aggregate the two into a single favourable summary. What replicates on both is the division of labour the decoupling predicts, with SLR-C moving the pointwise-fit metrics and gated UTD moving rank correlation.

Table 16 also reports four matched published label-distribution objectives, δ -LDL [24], LDL-LRR [22], LDL-DPA [23], and LDL-DVS [25], all of which stay essentially identical to the baseline under these strong frozen CLIP features, and Table R2 of this letter adds the seed dispersion and the per-metric p -values omitted from the manuscript for space.

Where and what we changed. the new Sec. 4.5, with the EMOTIC study in Sec. 4.5.1 and Table 15 and the label-distribution study in Sec. 4.5.2 with Table 16, the dataset descriptions and metrics in Sec. 4, and the added transfer limitations in Sec. 5.

Comment 2

More comparisons with state-of-the-art Large Language Model methods (such as 2026) are needed.

Response: *We agree, and add two 2026 intent-recognition methods plus four recent MLLMs; both tables are reproduced below.*

CoT4Intent (2026) probes hallucination when MLLMs are prompted with chain-of-thought reasoning for visual intention, and is now a row in Table 2 (49.54 Avg. F1). It bears directly on our design rationale, because it finds that MLLM-generated intent reasoning contains evidence that does not ground the predicted intent, which is why UTD consumes rationales through an agreement

gate rather than trusting them uniformly. We make this connection explicit in Related Work.

HVU-CLIP (TPAMI 2026) is the journal extension of IntCLIP by the same authors. Its published numbers (50.22 Avg. F1) enter Table 2 under its own CLIP ViT-B/32 backbone and are marked accordingly, since no code has been released. Because that backbone differs from ours, Table 14 adds a same-backbone comparison under ResNet101, where FDIL outperforms HVU-CLIP on three of the four commonly reported metrics, namely Macro F1 (43.20 vs. 41.24), Samples F1 (52.02 vs. 50.41), and Avg. F1 (48.62 vs. 47.96), while trailing on Micro F1 (50.63 vs. 52.22).

Zero-shot MLLMs. The zero-shot comparison now has its own table (Table 3), adding three 2026 open-weight models, namely Qwen3.5-9B, Step3-VL-10B, and Gemma-4-E4B-IT, plus GPT-5.6 Sol. The frontier model is genuinely strong, reaching 49.04 Avg. F1, while remaining 8.24 points below FDIL. We read this as evidence that stronger pretrained priors narrow the gap, while task-specific learning retains a clear advantage on aggregate fixed-threshold F1.

Where and what we changed. the two new works discussed individually in Sec. 2, the CoT4Intent and HVU-CLIP rows in Table 2, the same-ResNet101 comparison in Table 14, and the dedicated zero-shot comparison in Table 3.

[Manuscript Table 2] Main comparison on Intentionomy test split (excerpt: recent methods and ours). Avg. is the mean of Macro/Micro/Samples F1. [†]quoted under its own CLIP ViT-B/32 backbone because no code has been released; a backbone-matched comparison under ResNet101 is given in manuscript Table 14.

Method	Publication	Macro F1	Micro F1	Samples F1	Avg.
PIP-Net	TMM 2023	32.57	45.94	47.05	41.85
CoT4Intent	-	41.70	53.10	53.82	49.54
HVU-CLIP (ViT-B/32) [†]	TPAMI 2026	42.91	55.22	52.54	50.22
IntentMLM	PR 2026	43.05	54.77	56.75	51.52
IntCLIP (ViT-L/14)	ECCV 2024	44.98 $\pm$ 1.12	54.73 $\pm$ 0.49	54.73 $\pm$ 1.18	51.48 $\pm$ 0.82
HLEG (ViT-L/14)	TAC 2023	47.38 $\pm$ 0.75	54.85 $\pm$ 1.70	55.29 $\pm$ 1.06	52.50 $\pm$ 0.98
LabCR (ViT-L/14)	TIP 2024	47.87 $\pm$ 0.38	56.78 $\pm$ 0.97	57.23 $\pm$ 1.25	53.96 $\pm$ 0.71
CLIP ViT-L/14	ICML 2021	47.33 $\pm$ 1.15	54.98 $\pm$ 2.90	55.74 $\pm$ 2.64	52.68 $\pm$ 1.96
FDIL (UTD only)	Ours	49.90 $\pm$ 1.18	58.01 $\pm$ 1.03	58.43 $\pm$ 1.13	55.45 $\pm$ 1.02
FDIL	Ours	51.90 $\pm$ 0.95	59.90 $\pm$ 0.62	60.03 $\pm$ 0.94	57.28 $\pm$ 0.75

Comment 3

The paper lacks theoretical analysis or deeper learning insights. It must add more value to the paper if more theoretical analysis were conducted, e.g., from information-theoretical or feature selection perspectives.

Response: We agree, and rebuild Sec. 3.5 around exactly those two perspectives, with Sec. 4.3.2 testing what they predict.

Information-theoretic side. Eq. (15) treats the intent label as a random human judgement rather than a deterministic ground truth, giving the exact

[Manuscript Table 3] Zero-shot MLLM comparison on the Intentionomy test split.
[‡]Published result quoted from IntentMLM.

Model	Macro	Micro	Samples	Avg.
Zero-shot prompting				
Qwen3.5-9B [46]	37.11	36.50	36.40	36.67
Step3-VL-10B [47]	38.21	40.21	36.10	38.17
Qwen3-VL-8B-Instruct [48]	38.51	40.75	39.63	39.63
InternVL3-8B [49]	37.54	41.85	40.50	39.97
Gemma-4-E4B-IT [50]	38.76	42.44	39.25	40.15
GPT-4o [51] [‡]	39.48	40.57	41.45	40.50
GPT-5.6 Sol [52]	47.58	49.64	49.91	49.04
Task-trained reference				
CLIP ViT-L/14 [41]	47.33	54.98	55.74	52.68
FDIL (ours)	51.90	59.90	60.03	57.28

three-term identity $H(Y_c | B) = H(Y_c | X) + I(Y_c; X | Z) + I(Y_c; Z | B)$, which separates irreducible subjective disagreement, the intent-relevant information the frozen representation discards, and the information the baseline readout leaves unused. The middle term is unrecoverable downstream by the data-processing inequality, and we now say explicitly that FDIL does not address it. SLR-C’s contribution is stated as a conditional-information claim, $H(Y_c | B, S) = H(Y_c | B) - I(Y_c; S | B)$ (Eq. (16)), and because the prior is computed from the same frozen feature, this gain satisfies $I(Y_c; S | B) \leq I(Y_c; Z | B)$, so SLR-C addresses readout insufficiency only. The gain vanishes when the prior is redundant given the baseline scores; SLR-C’s improvement over the baseline is consistent with the prior supplying additional label-relevant information, and the prior ablation (Table 8), where no single source dominates, indicates that the textual sources contribute complementary content.

Feature-selection side. Eq. (17) formalizes the Top- K stage as candidate screening followed by margin correction. Inside the screened set, reranking shifts a confusable pair’s margin by exactly $\alpha(\tilde{s}_{c+}^{\text{prior}} - \tilde{s}_{c-}^{\text{prior}})$, and nothing outside it changes. In screening terms K trades coverage against distractor noise, and both ends are measured. The baseline Top-10 set covers 83.9% of positive labels, and the sensitivity study (Table 9) shows an interior optimum at $K = 10$.

UTD as reliability-weighted privileged information. The rationale teacher is now stated for what it is, a channel available in training and discarded at test time [27, 28]. That turns the gate into an estimation question with an answer, because the mean-squared-error-optimal weight $g^* = \sigma_v^2(\omega)/(\sigma_v^2(\omega) + \sigma_t^2 + b_t^2)$ of Eq. (18) is monotonically decreasing in agreement, so our $g = 1 - \omega$ is a parameter-free monotone approximation of it rather than an unstructured heuristic. We are careful about the status of that surrogate: since $\sigma_v^2(\omega)$, σ_t^2 , and b_t are not observable in practice, the manuscript presents $g = 1 - \omega$ as following Eq. (18) in trend rather than as the exact optimum.

The experiment these derivations predict. Sec. 4.3.2 stratifies samples by the two ambiguities and measures each module per cell, cross-fitted over the

train split because it is the only split carrying per-sample agreement. The semantic axis confirms the prediction cleanly. The supervisory axis is directionally consistent but weaker, and its low-ambiguity stratum holds only 1,859 images, so we claim no monotone trend. The full model also does not exceed its stronger component in the doubly-ambiguous cell. We therefore use the experiment to support the predicted localization of the two effects, not to claim a gain from composing them.

Where and what we changed. the derivations (15)–(18) and the module mapping in Sec. 3.5, the stratified analysis in Sec. 4.3.2 and Table 7, and the cross-dataset test in Sec. 4.5.2.

Comment 4

The overall framework is over-complicated and lacks clarity in design justification. Multiple modules are introduced, but their individual contributions and interactions are not clearly disentangled. The ablation studies, while present, are not sufficient to demonstrate that each component is necessary rather than redundant, making the method appear over-engineered.

Response: *FDIL has two functional modules, only one of which exists at inference, each one dictated by a distinct term of the decomposition; the 2×2 ambiguity-stratified analysis of Sec. 4.3.2 disentangles their individual contributions, and four further tests separate necessity from redundancy.*

On complexity. FDIL has two modules, a fixed semantic prior with a residual head (SLR-C) and a training-only text teacher (UTD). UTD’s effect is retained in the student’s parameters, but the teacher is discarded after training and adds no inference-time computation, so the deployed model is a lightweight residual MLP over frozen features at 2.41M parameters and 0.47 ms per image at the head level (Table 17).

On design justification. The two modules are not accumulated components but the two quantities the entropy decomposition of Eq. (15) leaves actionable. The readout-insufficiency term $I(Y_c; Z \mid B)$ is a property of the classifier head given a frozen representation, and SLR-C is the module that acts on it, by supplying conditional information from a fixed semantic prior (Eq. (16)) and correcting margins inside the screened candidate set (Eq. (17)). Unreliable annotation is instead a property of the supervision signal, not of the head, and is handled by treating rationales as privileged information whose optimal weight decreases in annotator agreement (Eq. (18)). Because the two act on different objects, one at inference on scores and one at training on targets, the decomposition motivates treating them with distinct mechanisms rather than forcing them into a shared target. This design choice is further supported empirically by the unified controls in Table 6.

On disentangling individual contributions. The 2×2 ambiguity-stratified analysis of Sec. 4.3.2 is added for this purpose. It stratifies samples into a 2×2

grid by the two ambiguities, semantic ambiguity by the negative top-2 margin of out-of-fold baseline logits and supervisory ambiguity by $1 - \omega$, and measures each module separately per cell against a shared baseline with paired t -tests and Holm correction over the twelve comparisons (Table 7, reproduced in the Reviewer 2 block above). The two modules localize to different cells. SLR-C gains only where semantic ambiguity is high (+4.08 and +1.17 F1, both significant) and nothing in the low-semantic cells, so its effect tracks the margin axis independently of annotator agreement. UTD is inert wherever agreement is high (-0.05 and -0.07 F1) and its only significant gain is the doubly-ambiguous cell (+0.84, corrected $p = 0.025$). These patterns are consistent with distinct functional sensitivities of the two modules. We also state what the same analysis does not support: in the doubly-ambiguous cell the full model tracks its stronger component (+1.14 against SLR-C’s +1.17) rather than exceeding it, so we claim the predicted localization of the two effects and not additive composition, and we note that the low-supervisory-ambiguity stratum holds only 1,859 images.

On necessity rather than redundancy. Beyond the stratified analysis, the manuscript offers four independent tests. The unified control shows that merging both signals into one distillation target performs no better than either alone (Table 6). Shuffling rationales across images is worse than removing UTD entirely (-10.40 against -4.47 Avg. F1), so the gain depends on image-specific content rather than on adding external text (Table 13). Removing or flattening the agreement gate costs -1.97 and -1.12 Avg. F1 and degrades every other aggregate metric, the one exception being that the uniform gate gains +1.13 on the hard subset (Table 13). And the two label-distribution transfers separate the modules by which metric each one moves, on both datasets, where they cannot share a fitted threshold. These distinct response patterns provide evidence against simple redundancy between the two components.

Where a component is not clearly necessary. FDIL’s advantage over its own UTD-only ablation does not reach significance on Samples F1 at $n = 5$ ($p_H = 0.0755$ in Table R1), although it does on Macro, Micro, Avg., mAP, and hard-subset F1. The case for the decoupled design therefore rests on the overall evidence together with modularity, since each function stays separately measurable and UTD can be deployed alone at zero additional inference cost.

Presentation and notation. The unified control and the stratified analysis are now one subsection arguing effectiveness and necessity together, and every results subsection is reduced to its table or figure, a short analysis, and a conclusion. The notation table is retained in Sec. 3 (Table 1) and reorganized into a compact two-column layout. Entries are now ordered row-wise by the module that introduces them, so that the problem setup, SLR-C, and UTD symbols form contiguous blocks, and the relation between the base logits $\mathbf{s}^b$ and the trainable student logits $\mathbf{s}^v$ is stated in the table itself. We also unify the symbols of the analysis in Sec. 3.5 with those of the method. The frozen encoder is now written f_{img} throughout instead of carrying a second name ϕ in the analysis, the image feature is named once as $\mathbf{z} = f_{\text{img}}(x)$ and reused by the semantic prior, the residual head, and the analysis, and the auxiliary map

ψ is removed. The random variables X, Y_c, Z, B, S , previously defined only in words at first occurrence, are listed in the table as the counterparts of x , the annotation of intent c , $\mathbf{z}$, $\mathbf{s}^b$, and $\tilde{\mathbf{s}}^{\text{prior}}$. Finally we remove three symbol collisions: the per-source normalization statistics are now $\text{mean}^r(x)$ and $\text{std}^r(x)$ rather than μ^r, σ^r , which freed $\sigma(\cdot)$ for the sigmoid alone, the teacher head is h_{tea} rather than g_{tea} , which freed g for the agreement gate alone, and the annotation and teacher noise terms are η_v, η_t rather than $\varepsilon_v, \varepsilon_t$, which freed ϵ for the numerical-stability constant alone.

Where and what we changed. the design rationale in Sec. 3.5, the stratified analysis in Sec. 4.3.2 with Table 7, Tables 6, 13, and 5, the dissociation in Sec. 4.5.2, the significance discussion in Sec. 4.2.1, and the modularity argument in Sec. 5.

[Manuscript Table 6] Unified versus functionally decoupled ambiguity handling. Results are means over three seeds.

Method	Macro	Micro	Samples	Avg.	Hard
Unified UTD only	50.20	58.03	58.79	55.68	30.27
Unified SLR-C only	51.29	60.10	59.22	56.87	34.01
Unified joint-target KD	50.69	58.35	57.96	55.66	32.41
FDIL	52.13	60.06	60.28	57.49	35.02

[Manuscript Table 13] Negative controls on the UTD text teacher (Δ in points; negative denotes degradation).

Condition	Macro	Micro	Samples	Avg.	mAP	Hard
FDIL (full)	52.13	60.06	60.28	57.49	55.10	35.02
w/o UTD	-4.23	-4.08	-5.10	-4.47	-4.93	-5.31
Shuffled rationale	-7.75	-12.39	-11.01	-10.40	-9.04	-7.58
Ungated teacher	-2.44	-2.16	-1.30	-1.97	-1.49	-3.33
Uniform agreement gate	-1.16	-0.93	-1.21	-1.12	-1.33	+1.13

Comment 5

The manuscript must require careful proofreading, such as [7] The manuscript must require careful proofreading and [9] Computer Vision - ECCV 2024.

Response: *Both entries are corrected, followed by a bibliography-wide pass.*

Entry [9] now reads *Computer Vision – ECCV 2024* with a proper en dash instead of the stray hyphen, and we re-check entry [7] together with all neighbouring entries. We then standardize venue names and abbreviations across the bibliography and verify the venue information of the remaining entries. We also re-read the manuscript for grammar, acronym consistency, and cross-reference correctness.

Where and what we changed. the corrected venue fields in the Bibliography including entry [9], the standardized venue names and abbrevia-

tions throughout, and the terminology and cross-reference corrections in the manuscript body.

5. Response to Reviewer 6

We thank the reviewer for the positive assessment of the writing and structure, and for the precise methodological requirement, which we implement in full.

Comment 1

The paper presents a functionally decoupled framework for visual-intent recognition that addresses supervisory and semantic ambiguity using structured semantic priors and uncertainty-aware distillation. The manuscript is well written, clearly structured, and the proposed contribution is reasonably good and relevant to the field. However, the performance improvement over comparable state-of-the-art methods is relatively small, which limits the strength of the empirical contribution. Although the authors report results using mean $\pm$ standard deviation over three runs, this alone is not sufficiently convincing for establishing statistical superiority. An explicit paired t -test, applied consistently against the strongest state-of-the-art methods, should therefore be provided to demonstrate that the observed improvements are statistically significant rather than caused by random variation.

Response: *We implement the requested test, namely five seeds with paired two-sided t -tests against every matched-protocol comparator, Holm-corrected, with the full statistics in Table R1 of this letter. We additionally strengthen the empirical contribution itself, with a cross-dataset study and an extended comparison against two 2026 methods and four recent multimodal large models.*

We agree that mean $\pm$ std over three runs does not establish statistical superiority.

Protocol. We extend from three to **five** independent seeds. For each seed we retrain, on that same seed, the matched-protocol reproductions of the three strongest comparable methods, namely HLEG, LabCR, and IntCLIP, over identical frozen CLIP ViT-L/14 features, the identical split, and identical validation-only class-wise thresholding. This gives five *paired* observations per method, with the training run as the pairing unit, so a paired two-sided t -test on the per-seed differences is well defined. Since FDIL is tested against five methods per metric, we also report Holm-corrected p -values.

Outcome. Avg. F1, mAP, and hard-subset F1 are significant against every matched-protocol comparator after correction, the published-only rows being quoted from their papers and hence outside the paired design. Against the strongest compared method, LabCR, the gains are +3.32 Avg. F1 and +7.67 hard-subset F1. The class-wise-thresholded F1 metrics carry more seed-to-seed

variance because thresholds are fitted on a 498-image validation split, and the only non-significant contrast is Samples F1 against our own UTD-only ablation ($p_H = 0.0755$), which we report alongside the significant ones. The gains over the matched published methods are therefore statistically reliable, while remaining consistent with the modest effect size the reviewer correctly identified. Sec. 4.2.1 states the outcome and protocol, and the full statistics are provided in Table R1.

Beyond the requested test. Because the reviewer’s concern is ultimately about the strength of the empirical contribution and not only about its statistical treatment, we also broaden the evidence in two ways. First, the evaluation is no longer confined to a single benchmark: the new Sec. 4.5 adds a cross-dataset study, comprising a controlled 5×5 cross-validation on the larger EMOTIC benchmark, where every aggregate metric improves significantly over the matched baseline, and a transfer to two label-distribution benchmarks, Flickr-LDL and Emotion6, where the class-wise threshold search disappears entirely and the division of labour between the two modules nevertheless replicates. Second, the comparison set is extended with two 2026 intent-recognition methods, CoT4Intent and HVU-CLIP (TPAMI 2026), the latter also under a same-backbone ResNet101 comparison, together with four recent multimodal large models, Qwen3.5-9B, Step3-VL-10B, Gemma-4-E4B-IT, and GPT-5.6 Sol, in a dedicated zero-shot table. Together with the paired tests above, this places the reported improvements against both stronger baselines and unseen datasets rather than against seed variation alone.

Where and what we changed. Sec. 4.2.1, the confidence intervals and re-generated matched-protocol rows in Table 2 whose caption now states five seeds, the new Sec. 4.5 with Tables 15 and 16, the new rows in Tables 2, 14, and 3, and Table R1 of this letter.

6. Response to Editor-in-Chief Checklist

We check the revision against each item and report the outcome below.

Comment 1

Take a careful look at your bibliography and they cover the state of the art. Missing references from last and current year most probably would mean you are missing the state of the art and the revision process can be delayed being asked to update it. Please do not make excessive citation to arXiv papers, but substitute them with their peer-reviewed versions, or papers from a single conference series. Do not cite large groups of papers without individually commenting on them. So we discourage “In prior work [1,2,3,4,5,6] ...”. Your bibliography in the final version after the revision still should be between 35-55 items.

Response: *Recent coverage extended, all new works discussed individually, 52 cited items.*

Recent coverage. We add two 2026 intent-recognition works, CoT4Intent and HVU-CLIP (TPAMI 2026), alongside the IntCLIP (ECCV 2024) and IntentMLM (Pattern Recognition 2026) references already present. We further compare against four 2026 multimodal large language models, Qwen3.5, Step3-VL, Gemma-4, and GPT-5.6 Sol, and add the latest label-distribution-learning work, LDL-DVS (ICML 2026).

Remaining items. Every entry is cited in its peer-reviewed version wherever one exists, each newly added method is discussed individually by name and contribution in Sec. 2 rather than appended to a bracket list, and the compiled manuscript contains 52 cited items, within the required 35–55 range.

Where and what we changed. the new entries and corrected venue fields in the Bibliography, and the individually discussed additions in Sec. 2.

Comment 2

Please make sure the revised version is relevant to the readership of the Pattern Recognition field. To this end, please make sure you cite RECENT work from the field of pattern recognition not only the Pattern Recognition journal.

Response: *The bibliography spans the recent pattern-recognition literature rather than a single venue.*

Cited venues include IEEE TPAMI (HVU-CLIP, 2026), Pattern Recognition (IntentMLM, 2026), ICML 2026 (LDL-DVS), ICML 2025 (δ -LDL), IEEE TKDE (LDL-LRR, 2023), IEEE TNNLS (LDL-DPA, 2024), IEEE TIP (LabCR 2024, CPAD 2023), IEEE TAC (HLEG 2023), IEEE TMM (PIP-Net 2023), and ECCV 2024 (IntCLIP), together with the multi-label classification and noisy-label lines from CVPR, NeurIPS, ICML, ICLR, and AAAI.

The problem itself, multi-label recognition under ambiguous and subjective supervision with calibration and threshold selection treated explicitly, sits squarely within the journal’s scope.

Where and what we changed. the venue distribution of the Bibliography, and the multi-label and label-noise method families compared in Table 2.

Comment 3

Although the revision could lead to extending your article, it still can not exceed the page limits or violate the format, i.e. double spaced SINGLE column with a maximum of 35 pages for a regular paper and 40 pages for a review.

Response: *The compiled marked-up manuscript is 35 pages, double-spaced and single-column, exactly at the limit.*

The revised manuscript is double-spaced and single-column throughout. To fit the newly requested evidence within the limit, we condense repeated protocol text and combine tables where possible, and several supplementary diagnostic tables therefore appear in this letter instead of the manuscript. The compiled review manuscript is **35 pages**, within the page limit.

Where and what we changed. the 35-page compiled review manuscript and its unchanged double-spaced, single-column format.

7. Closing Statement

We thank the Associate Editor and all reviewers for a careful and constructive second round. The two concerns the Associate Editor emphasized, statistical validation and evaluation breadth, drive substantive new work. We now report five-seed training with confidence intervals and paired t -tests against every matched-protocol comparator, together with a cross-dataset study placed in the manuscript itself, covering the larger EMOTIC benchmark and the Flickr-LDL and Emotion6 label-distribution benchmarks.

The submitted marked-up manuscript remains double-spaced and single-column and compiles to exactly **35 pages**. The three supplementary tables that did not fit are reproduced in this letter, alongside the manuscript tables and the case-study figure the reviewers asked about.

We thank the Associate Editor and reviewers again for their careful guidance and hope that the revised manuscript now addresses all concerns.